

Background:

I have always been fascinated by satellite imagery and the way large-scale landscapes can be understood through patterns rather than individual measurements. For this project, I wanted to explore a place that is personally meaningful to me: Sambhar Lake in Rajasthan, India. Sambhar Lake is a saline, ephemeral lake whose surface water extent changes noticeably with seasonal monsoon rainfall. Rather than focusing on precise hydrological modelling, this study aims to visually explore how surface water patterns vary across years and whether distinct spatial zones consistently emerge within the lake.

Table 1-1

year	prep_total_m m	tavg_mean_C	tmean_mean _C	prep_MAM_ mm	tavg_MAM_ mean_C	tmean_MAM_ mean_C
2022	33.98	7.921568627	8.096341463	0.69	8.994047619	9.227142857
2023	23.63	7.835616438	7.816257669	4.49	8.280434783	8.311538462
2024	47.03	8.148333333	8.094545455	0.1	8.750574713	8.560294118

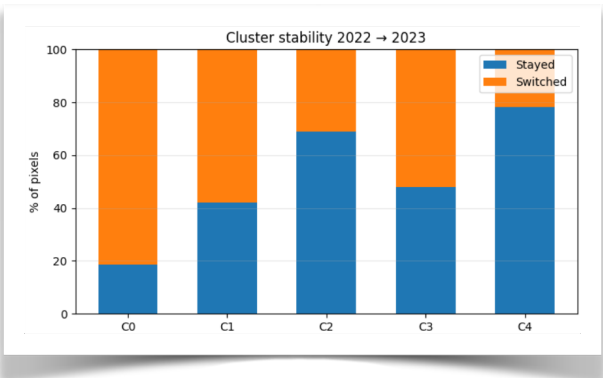
Data:

This analysis uses multispectral Sentinel-2 imagery provided by the European Space Agency through the Copernicus Programme. One cloud-free image from the post-monsoon period was selected for each year (2022–2024). Annual precipitation data were obtained from a pre-aggregated climate summary dataset (Table 1-1) covering the Sambhar region, providing total yearly rainfall values for comparison with surface water patterns.

Methods:

Green (B03) and near-infrared (B08) Sentinel-2 bands were used to compute the Normalised Difference Water Index (NDWI), which highlights surface water presence by contrasting water-absorbing and water-reflecting wavelengths. NDWI rasters were clipped to the Sambhar Lake boundary to focus analysis strictly within the lake extent.

To explore spatial structure, unsupervised k-means clustering (k = 5) was applied separately to each year’s NDWI raster. This approach groups pixels with similar NDWI values without predefined labels, allowing dominant surface water regimes to emerge naturally from the data. Clusters were relabelled based on relative NDWI magnitude to enable comparison across years.



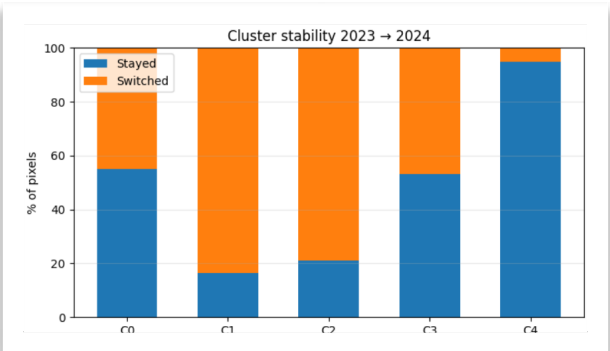
Graph illustrating comparison of clusters in the year 2022 and 2023

Scope and Interpretation Note:

This analysis is exploratory and based on remotely sensed surface indicators rather than in situ hydrological or geochemical measurements. The clusters identified through NDWI-based k-means clustering are interpreted qualitatively, using relative

NDWI magnitude, spatial location, and inter annual behaviour. While distinct wet,

transitional, and dry surface regimes are expected in saline lake systems, the clusters presented here are not intended as definitive geological or compositional classifications.



Graph illustrating comparison of clusters in the year 2023 and 2024

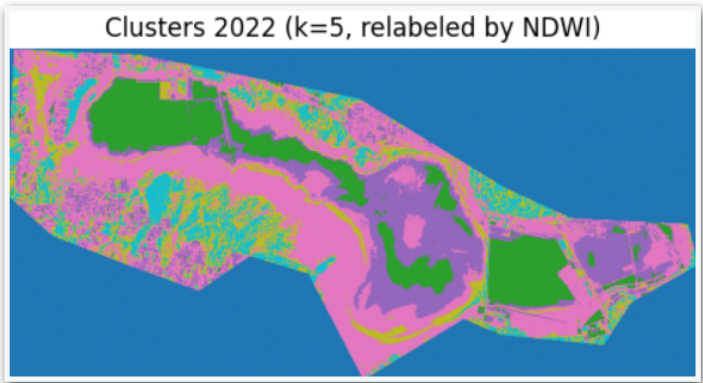
Results:

The NDWI maps reveal clear spatial variability in surface water presence across Sambhar Lake, with noticeable inter-annual differences. Clustering highlights distinct zones ranging from persistently wet regions to consistently dry, high-reflectance areas. Cluster-rainfall comparisons show that years with higher precipitation generally correspond to increased NDWI values, though the magnitude of response varies by cluster. Transition analyses indicate that some clusters remain spatially stable across years, while others exhibit frequent

switching, particularly in intermediate zones.

Interpretation:

Clusters with consistently high NDWI values tend to remain spatially stable, suggesting persistent water-retaining regions within the lake. In contrast, clusters with intermediate NDWI values show greater inter-annual variability and higher transition rates, indicating sensitivity to rainfall fluctuations. Drier clusters display limited responsiveness to precipitation, likely reflecting salt-crust or evaporative zones. These patterns suggest that Sambhar Lake is composed of spatial regimes that respond differently to monsoon variability rather than behaving as a uniform surface.



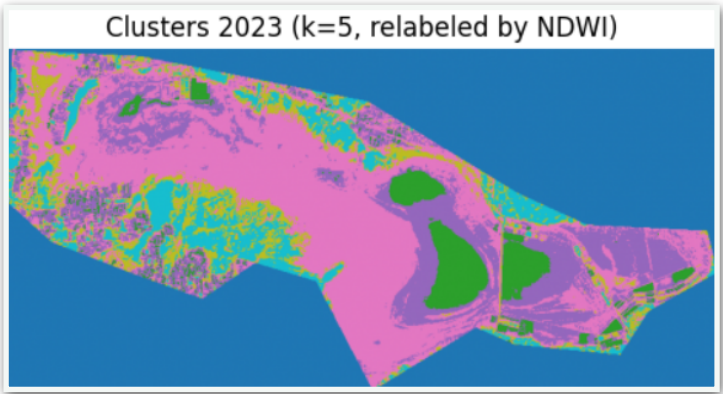
Colour (as seen on map)	Cluster ID	Relative NDWI	Colour tag	Expected surface interpretation
Dark Green	Cluster 0	Very high	Persistent Wet Core	Stable, water-retaining or low-lying zones
Light Cyan / Teal	Cluster 1	Moderately high	Seasonally Inundated Fringe	Shallow or periodically flooded regions
Yellow/Olive	Cluster 2	Intermediate	Transitional Moist Zone	Rain-sensitive surfaces with high variability
Pink / Magenta	Cluster 3	Low	Ephemeral Drying Areas	Temporarily wet surfaces undergoing drying

Purple / Violet	Cluster 4	Very low	Persistent Dry / Salt-Crust Zone	Highly evaporative or salt-dominated surfaces
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Clusters ($k = 5$) were derived from Sentinel-2 NDWI and relabelled by relative NDWI magnitude. Colours indicate surface regimes from persistently wet (green) to persistently dry or salt-crust dominated (purple), with intermediate zones showing greater variability. This classification is exploratory and reflects surface conditions inferred from remote sensing rather than direct hydrological or compositional measurements.

Discussion:

Previous geochemical and sedimentological studies describe Sambhar Lake as a spatially heterogeneous evaporative system shaped by long-term hydrological and climatic processes. The clustering patterns observed here align with this understanding, providing a surface-level visualisation of variability that complements earlier subsurface and mineralogical research. While NDWI does not directly measure water depth or salinity, it offers an accessible proxy for exploring spatial water dynamics. This study emphasises visualisation and pattern recognition rather than causal inference. In saline lake environments, surface reflectance can also be influenced by biological activity such as algal or microbial blooms, which may affect NDWI values. As a result, the observed clusters likely reflect combined surface processes rather than water presence alone, reinforcing the exploratory nature of this analysis.



Conclusion:

This project demonstrates that NDWI-based clustering can effectively reveal structured yet dynamic surface water patterns in Sambhar Lake across multiple years. By combining satellite imagery, simple unsupervised learning, and climate context, the analysis highlights how visual exploration can support intuitive understanding of complex environmental systems. Such lightweight spatial analyses can serve as a useful starting point for deeper hydrological or geochemical investigations.

